

# STELLAR ASTROPHYSICS

Astronomy 310, Fall 2026

Department of Astronomy, University of Wisconsin–Madison

## Course Information

**Course Description:** Properties of normal and peculiar stars as found from an analysis of the radiation they emit; introduction to radiation transfer. Theory of stellar atmospheres, interiors, and evolution.

Stars are the building blocks of the Universe. They drive the energetic and chemical evolution of galaxies, and the elements heavier than helium that make up the world around us were forged inside them. This course develops the physics of stars in the order an astronomer encounters them: first what we observe, then how we model the light that reaches us, and finally what is happening inside. We begin with stellar observations (distances, magnitudes, colors, spectra, binaries, the Hertzsprung–Russell diagram) and the parameters they yield. We then study stellar atmospheres, where radiation and matter interact to produce the spectra we measure: the radiative transfer equation, opacity sources, the Eddington gray-atmosphere approximation, and the formation of spectral lines. With those observational and atmospheric foundations in place, we turn to stellar interiors and the equations of stellar structure: hydrostatic equilibrium, the equation of state including electron degeneracy, energy transport by radiation and convection, and nuclear energy generation from the pp chain through advanced burning. We close with a more compact treatment of stellar evolution, surveying the main sequence, post-main-sequence phases, stellar remnants (white dwarfs, neutron stars, black holes), supernovae, and close-binary evolution.

**Credits:** 3 (Carnegie Unit)

**Credit Hour Explanation:** This class meets for two, 75-minute class periods each week. For each credit earned, students should expect to spend approximately two hours per week outside of class completing readings, assignments, studying, and other course activities. Thus, for this 3-credit course, students should anticipate about 6 hours of out-of-class work per week, in addition to scheduled class meetings.

**Requisites:** MATH 222 and (PHYSICS 205, 241, or 249, or concurrent enrollment).

## Course Designations/Attributes:

- Breadth: Physical Science. Counts toward the Natural Science requirement.
- Level: Intermediate
- L&S Credit: Counts as Liberal Arts and Science credit in L&S
- Repeatable for Credit: No

## Course Learning Outcomes

By the end of this course, students will be able to:

1. Describe the properties of stars and explain how these properties are deduced from the radiation that they emit.
2. Define the basic principles of radiative transfer and apply these principles to interpreting the atmospheres of stars.
3. Apply basic physical principles and mathematical techniques learned in prerequisite classes to understand the structure, evolution, and ultimate fates of stars.
4. Demonstrate thinking like a scientist, applying heuristics such as dimensional analysis and order-of-magnitude estimation to obtain quick insights into unfamiliar systems.
5. Communicate scientific concepts and results clearly, using a combination of writing and visual representations with bibliographic references as appropriate.
6. Access, read, and critically evaluate existing scientific literature and information repositories relating to stellar astrophysics.

**Meeting & Instructor Information**

|                   |                        |                      |                                                          |
|-------------------|------------------------|----------------------|----------------------------------------------------------|
| <b>Lecture:</b>   | Mon & Wed 2:30–3:45 PM | <b>Location:</b>     | Sterling Hall, Room 1333                                 |
| <b>Professor:</b> | Melinda Soares-Furtado | <b>Email:</b>        | <a href="mailto:mmsoares@wisc.edu">mmsoares@wisc.edu</a> |
|                   |                        | <b>Office:</b>       | Chamberlin Hall, Room 6275                               |
|                   |                        | <b>Office Hours:</b> | Mon 9:30–11:00 AM; Wed 4:00–5:30 PM                      |

**Materials**

**Link to Canvas Site:** This course will be administered through Canvas, with additional course information available on the course website: <https://msoaresfurtado.com/Courses.html>. Course readings and assignment information will be posted there, while all assignments should be submitted via file upload on the Canvas site.

**Other Digital Instructional Tools:** Piazza will be the primary form of communication for this course.

**Required Textbook:**

- o Pinsonneault, M.H., and Ryden, B.S. *Stellar Structure and Evolution* (Cambridge University Press, 2023; ISBN 978-1-009-19384-3). [\[link\]](#)

Chapter readings are listed in the course outline at the end of this syllabus.

**Software:** Calculating realistic stellar structure and evolution models requires a numerical code. For one of the homework assignments we will use (Modules for Experiments in Stellar Astrophysics) MESA Web, an online interface to the MESA code that runs in your browser and requires no installation. MESA Web is freely available at <https://mesastar.org/>. Full instructions on how to use the interface for our problem sets will be provided in class.

**Coursework & Grading**

Grades are based on points earned from homework, weekly online reflection questions, and one final exam. Seven problem sets are assigned across the semester. Problem sets and reflection questions are submitted through Canvas. Grades are not curved, though AB/BC designations apply near boundaries. Attendance is not graded directly, but the weekly reflection questions described below serve as the class-participation component of the grade.

**Grading Policy:**

|                            |     |
|----------------------------|-----|
| Homework Sets .....        | 70% |
| Final Exam .....           | 25% |
| Reflection Questions ..... | 5%  |

**Grading Distribution:**

|         |         |
|---------|---------|
| A ..... | 90–100% |
| B ..... | 80–90%  |
| C ..... | 70–80%  |
| D ..... | 60–70%  |
| F ..... | <60%    |

**Problem Sets:** Seven problem sets are assigned and should be submitted on Canvas (see the course outline on page 3 for due dates). One of the seven sets includes a MESA Web computational component. Not every problem will be graded in detail: each set will have two to three “anchor” problems graded carefully for correctness and physical reasoning, with the remaining problems graded for completion and engagement. Anchor problems are not identified in advance.

Collaboration on problem-set strategy is welcome and encouraged: talking through approaches with classmates, comparing answers, and helping each other find errors are all normal and beneficial parts of doing physics. However, the final written work you submit must be your own, in your own words, and reflect your own understanding. If you collaborated with classmates on a problem, list their names at the top of the submission; this is not penalized and is good scholarly practice. Late problem sets lose 10% per 24 hours late, up to

72 hours; beyond that, late work is accepted only with a documented excuse. Each student is allowed one no-questions-asked 48-hour extension per semester, requested by email before the deadline.

**Reflection Questions:** Each week you will complete a short set of online reflection questions on Canvas, due Sundays at 11:59 PM. The prompts may ask you to summarize the week's central physical idea in your own words, identify which concept or derivation you found most challenging, or connect the week's material to topics covered earlier in the semester or to a current question in stellar astrophysics. Reflections are short (a few sentences per prompt is sufficient) and are graded on completion and engagement rather than correctness. Their purpose is to support active learning and to give the instructor ongoing feedback on what is and is not landing in lecture.

**Exams:** There will be one midterm and one final, both in-person and on paper. Each exam is one hour in duration. Both exams are open book: you may bring either required textbook (or a printout of relevant chapters), the recommended reference, your own handwritten or typed notes, and graded problem sets. Cell phones and laptops are prohibited; bring your own non-networked calculator. Exam questions will emphasize physical reasoning rather than memorized facts, so having reference material on hand will be useful but is not a substitute for understanding the material.

**Note on the Final Exam:** The registrar has scheduled our final exam for Wednesday, December 16, 7:45–9:45 AM. Because that is a very early hour, students are offered the option to take the final exam in class on the last day of class instead (Wednesday, December 9, 2:30–3:30 PM). You may choose either slot. Please declare your preference no later than the start of the last week of class (Monday, December 7) by submitting the preference form on Canvas. The final exam is one hour in length regardless of which slot you choose.

**Exam Dates:**

|                                       |                                   |
|---------------------------------------|-----------------------------------|
| Final Exam, Option 1 (in class)       | ..... Dec 9, 2026 (2:30–3:30 PM)  |
| Final Exam, Option 2 (registrar slot) | ..... Dec 16, 2026 (7:45–8:45 AM) |

## Academic Policies & Statements

**Accommodations for Students with Disabilities:** Faculty and students share responsibility in the accommodation process. Students are expected to inform faculty of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. I can work directly with you or in coordination with the McBurney Center to provide reasonable instructional and course-related accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA. <http://mcburney.wisc.edu/facstaffother/faculty/syllabus.php>

**Academic Honesty:** I expect you to do your own work, represent your ideas accurately, and give credit to others when you use their words or research. Discussing concepts with classmates is encouraged, but anything you submit must reflect your own understanding and writing. Use of online answer banks (e.g., Chegg, CourseHero) is not permitted. To support genuine learning, **please do not use AI tools (such as ChatGPT, Claude, or similar programs) to generate answers or complete assignments.** These tools can short-circuit your own reasoning process and prevent you from fully engaging with the material. Any academic dishonesty will result in a grade of zero on that specific assignment, and applicable University policies will be followed (<https://conduct.students.wisc.edu>). Please reach out to me if you are ever unsure about what is acceptable.

## Course Schedule

### Part I. Observations

#### Week 1 (Sep. 2)

Properties of stars: course overview, observing the Sun, and observed properties of stars.

*Reading: Chapter 1.1–1.2 (17 pp).*

#### Week 2 (Sep. 9)

Properties of stars: correlations among stellar properties, star clusters, and the H–R diagram.

*Reading: Chapter 1.3–1.4 (9 pp).*

**Homework 1 due Sep. 11.**

#### Week 3 (Sep. 14, 16)

Equations of stellar structure: conservation of mass, hydrostatic equilibrium, thermal equilibrium, and energy transport.

*Reading: Chapter 2.1–2.4 (11 pp).*

### Part II. Stellar Structure

#### Week 4 (Sep. 21, 23)

Equations of state: central pressure, quantum statistics, and ideal-gas pressure.

*Reading: Chapter 3.1–3.3 (9 pp).*

**Homework 2 due Sep. 25.**

#### Week 5 (Sep. 28, 30)

Equations of state and stellar energy transport: electron degeneracy, radiation pressure, and opacity.

*Reading: Chapter 3.4–3.5; Chapter 4.1 (20 pp).*

#### Week 6 (Oct. 5, 7)

Stellar energy transport: convection, mixing-length theory, and convective overshoot.

*Reading: Chapter 4.2–4.4 (12 pp).*

**Homework 3 due Oct. 9.**

#### Week 7 (Oct. 12, 14)

Stars as fusion reactors: quantum tunneling, energy generation and composition change, and the pp chain.

*Reading: Chapter 5.1–5.3 (18 pp).*

#### Week 8 (Oct. 19, 21)

Stars as fusion reactors and main-sequence stars: the CNO cycle, the triple-alpha process, internal structure, and polytropes.

*Reading: Chapter 5.4–5.5; Chapter 6.1–6.2 (15 pp).*

**Homework 4 due Oct. 23.**

#### Week 9 (Oct. 26, 28)

Main-sequence stars: homology, solar models, and zero-age main-sequence models.

*Reading: Chapter 6.3–6.5 (18 pp).*

### Part III. Stellar Evolution

#### Week 10 (Nov. 2, 4)

Star formation: molecular clouds and gravitational instability, isothermal collapse and fragmentation, and protostars.

*Reading: Chapter 7.1–7.3 (11 pp).*

**Homework 5 due Nov. 6.**

#### Week 11 (Nov. 9, 11)

Star formation: pre-main-sequence stars, the birthline and deuterium burning, and the initial mass function.

*Reading: Chapter 7.4–7.6 (13 pp).*

#### Week 12 (Nov. 16, 18)

Evolved stars: building a helium core, the Schönberg–Chandrasekhar limit, and shell hydrogen burning and red giants.

*Reading: Chapter 8.1–8.3 (14 pp).*

**Homework 6 (MESA Web) due Nov. 20.**

**Week 13 (Nov. 23, 25)**

Evolved stars: helium ignition and core helium burning, the asymptotic giant branch, s-process nucleosynthesis, superwinds, and planetary nebulae.

*Reading: Chapter 8.4–8.7 (17 pp).*

**Thanksgiving recess Nov. 26–29.**

**Week 14 (Nov. 30, Dec. 2)**

Ex-stars: white dwarfs, neutron stars and black holes, core-collapse supernovae, and r-process nucleosynthesis.

*Reading: Chapter 9.1–9.4 (19 pp).*

**Homework 7 due Dec. 4.**

**Week 15 (Dec. 7, 9)**

Course synthesis and review.

**Comprehensive Final Exam (Dec. 9).**